

Displaced Diffusion Binomial Tree for Real Option Valuation

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Abstract

This paper describes a real options valuation method for situations where the underlying asset may have negative values and the underlying project's present value distribution is something of the shape between normal and lognormal distribution, causing skewness to the rate of return distribution. The underlying project value is assumed to follow a displaced diffusion process with a dynamic path having up and down movements with properties of both additive and multiplicative processes. Therefore, the approach takes into account the changing volatility according to the level of the underlying asset value. A recombining binomial tree solution for the displaced diffusion is presented with a managerially applicable simulation-based parameterization. The results suggest that the assumptions of geometric Brownian motion and a lognormal distribution may not be sufficient for practical cash flow simulation-based real option valuation.

JEL Classification: G31, G13, D81

Keywords: Real options, binomial tree, displaced lognormal asset dynamics, sequential investments, value under uncertainty

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Extended Binomial Tree Valuation when the Underlying Asset Distribution is Shifted Lognormal with Higher Moments.

1. Introduction

Real options analysis (ROA) is a fairly new approach for valuing managerial flexibility in investment decisions. It has been suggested as an alternative valuation approach to calculate traditional net present value for situations where managerial flexibility and decisions under uncertainty can increase the project value in comparison with the static expected value assumed by net present value (NPV). These decisions can be related, for example, to deferring, extending, or abandoning the project. According to real options thinking, investments are sequential and irreversible, and made under conditions of uncertainty (Dixit and Pindyck 1995). The expression real options stems from utilizing and adapting the mathematics commonly used in valuing financial options related to some uncertain real investments. According to Trigeorgis and Mason (1987) and Brealey and Myers (2003), real options analysis can also be illustrated as an improved, economically corrected version of decision-tree analysis that has adopted the market perspective, allowing the determination of expected values using risk-neutral probabilities and a risk-free discounting rate. On the other hand, real options are also often used more as a metaphor for management philosophy of uncertainty and flexibility behind the investments (Lander and Pinches 1998; Thurner 2003). However, there are also deficiencies related to real options analysis. These include mathematical complexity, parameter estimations, lack of managerial skills to perform corrective actions during a project when needed, and the problems derived from the fact that models well suited to financial options are not necessarily suitable for real investments valuation (Lander and Pinches 1998; Busby and Pitts 1997; Miller and Waller 2003; Borison 2005; Triantis 2005).

Copeland and Antikarov (2001) proposed an alternative to overcome the earlier problems. They presented a whole valuation approach for real options analysis that combines several previous methods into a managerially understandable real option valuation framework. Their approach uses simulation of the present value cash flow calculation to estimate both the expected underlying asset value and its volatility. Then, real options are recognized and modeled with binomial trees. Copeland and Antikarov (2001), henceforth C&A, justify the use of the expected cash flow as the underlying asset with the so-called marketed asset disclaimer. The consolidated volatility estimation approach is based on the logarithmic present value that unifies all the different uncertainties into a single-parameter uncertainty measure.

This paper describes a real options valuation method for situations in which the underlying asset may have negative values and the underlying project's present value distribution is something of the shape between normal and lognormal distribution. The underlying project value is assumed to follow a displaced diffusion process with a dynamic path having up and down movements with properties of both additive and multiplicative processes. Therefore, the approach takes into account changing volatility according to the level of the underlying asset value. The valuation approach can be considered an extension of the real option valuation framework of C&A. However, this improved approach uses the displaced diffusion process, allows negative underlying asset values, and corrects the original volatility overestimation error in *ibid*. The method is also presented with a managerially applicable cash flow simulation-based parameterization. The results presented in this paper show that the assumptions of geometric Brownian motion and lognormal distribution may not be sufficient for practical real option valuation.

In Section 2 we present the real option valuation framework of Copeland and Antikarov (2001) and the related theoretical assumptions of the marketed asset disclaimer and consolidated volatility approach. In Section 3 we discuss generalized risk-neutral volatility estimation for lognormal cases. Next, in Section 4 we discuss the most common underlying asset dynamics. The following section 5 presents a generalized displaced diffusion process in the context of real options. In Section 6 we present modeling of the displaced lognormal binomial tree with a parameterization based on a cash flow simulation and a generalized risk-neutral volatility estimation. In Section 7 we use a case example adapted from a multi-stage investment project with several interacting real options presenting how the displaced diffusion assumption may change the results compared with the common assumption of geometric Brownian motion. Finally, the discussion summarizes this paper.

2. Real option valuation framework

Most real option articles focus mainly on how the valuation method works but they do not discuss the valuation process as a whole. While the seminal book of Trigeorgis (1996) was the first to introduce real options to a wider audience, the book of Copeland and Antikarov (2001) can be considered the first book to provide a hands-on approach for applied real option valuation. The book considers the valuation process as a whole, but also presents the idea of a consolidated valuation approach based on a Marketed Asset Disclaimer (MAD).

Instead of searching for comparable cases in financial markets for volatility estimation, C&A recommend using the present value of the project itself, without flexibility, as the underlying risky asset. The basic idea is to use the present value of the project's gross present value (GPV, present value of operating cash flows less investment outlays) as the underlying asset *without* the need for a twin security. This is regarded as the best unbiased estimate of a project's market value if it were a traded asset. This assumption is called the Marketed Asset Disclaimer (MAD), and is the same assumption used for NPV valuation. The results may be imprecise, especially when compared to the accuracy of many financial option models, but it is often the best that can be done. Also, other authors (e.g. Trigeorgis and Mason 1987; Trigeorgis 1996, p. 123) have presented similar ideas, and despite some justifiable criticism (e.g. Borison 2005), several papers on real options, including this one, use the same assumptions.

The first step in the valuation framework and process (Figure 1) is to calculate the gross present value – the net present value of the project less the investment costs. The discount rate used can be the weighted average cost of capital or some other justified equilibrium-based discounting rate. This is assumed to represent the project's underlying asset value without flexibility. The GPV calculation is not done with simple one-point estimates for different parameters. The most significant and uncertain values are modeled with distributions instead (such as triangle, PERT, normal and log normal distributions). Also, different serial and cross correlations between the parameters should be taken into account. The values for the single-point parameters, distributions, and econometric series can be defined by subjective managerial assumptions or from historical market data. The more market-based data, knowledge, and information there is the more reliable the estimates will be.

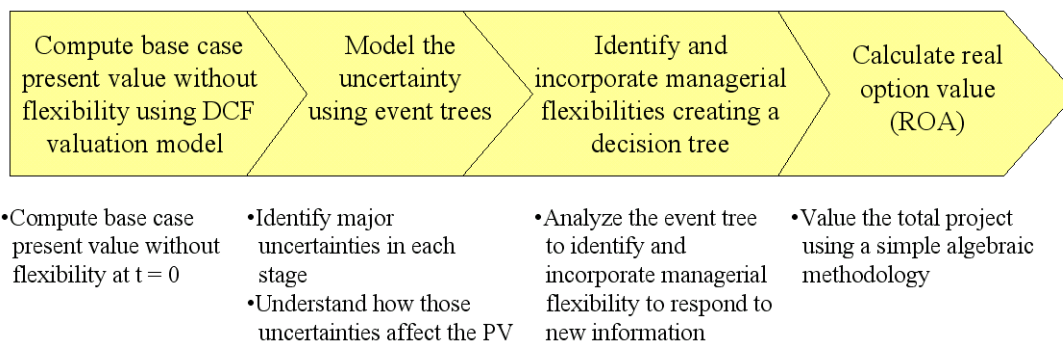


Figure 1: Real option valuation framework and process according to Copeland and Antikarov (2001).

In the second step, the uncertainty through time is modeled using Monte Carlo simulation with a consolidated logarithmic volatility approach. It is very difficult to analyze a lattice that is driven by more than two sources of uncertainty. To avoid this complexity, C&A present a consolidated volatility approach, which is based on two assumptions. The first is the use of a marketed asset disclaimer. The second, attributable to proof of Paul Samuelson (1965), is that the rate of return on any security, the properly anticipated price, and the cash flow will be a random walk, regardless of the pattern of cash flows (i.e. mean-reverting or cyclical) in the future, as long as investors have complete information about those cash flows. This enables combining any number of uncertainties into a spreadsheet by using Monte Carlo simulation techniques, and to produce an estimate of the present value and volatility of a project conditional on the set of random variables drawn from their underlying distributions. This consolidated volatility approach allows the reduction of multiple sources of uncertainty to only one unified volatility measure.

The idea of C&A's logarithmic present value approach is, similarly to most other consolidated volatility approaches, analogous to stock price simulations where the theoretical stock price is the sum of all future dividend cash payments, and with real options, these cash payments are the free cash flows. The sum of free cash flows' present value PV_0 at time zero is the current stock price (asset value), and at time one, the stock price PV_1 in the future. As the stock price at time zero is known while the future stock price is uncertain, only the uncertain future stock price is simulated. As a result, C&A use Monte Carlo simulation on the project's present value to develop a hypothetical distribution of one period returns. On each simulation trial run, the value of future cash flows is estimated at two time periods, one for the first time period and another for the present time. The cash flows are discounted and summed to the time 0 and 1, and their logarithmic ratio is calculated.

In the second step, when the volatility has been estimated, the event tree is built to model the uncertainty and dynamic behavior of the underlying value. It describes how much the value is likely to move up and down during the investment period. This is usually done with the binomial tree of Cox, Ross, and Rubinstein (1979), the CRR tree, and under the assumption that the project value follows stochastic geometric Brownian motion (gBm). This is also the most common assumption in real options research. The third step is to identify and incorporate managerial flexibility into the event tree. In the fourth step, the real options

analysis is conducted using simple algebraic (binomial) methodology with back-rolling and risk-neutral valuation.

The displaced diffusion binomial tree with its parameterization shares many similarities with Copeland and Antikarov. Both share the idea of using Monte Carlo simulated cash flow as the underlying asset according to the marketed asset disclaimer. Also, the idea of consolidating holds as well as the assumption of sufficiently smooth stochastic process without sudden significant jumps in the underlying asset value. However, a different stochastic process – displaced diffusion process – is assumed instead of geometric Brownian motion. Also, a volatility estimation approach is presented that corrects the error in the original C&A, and this approach is then extended for the displaced diffusion case.¹ Also, the CRR lognormal binomial tree is replaced with a displaced binomial tree that is based on displacing the equal probability binomial tree model of Wilmott et al. (1995).

3. Generalized risk-neutral volatility estimation

Although the fundamental idea in the approach of Copeland and Antikarov (2001) is correct, it has one technical deficiency. The method would be an appropriate estimation of volatility if the present value at time period 1 (PV_1) were period 1's expected (NPV) of subsequent cash flows, and this volatility would reflect the resolution of a single year's uncertainty and its impact on expectations for future cash flows. However, in C&A's solution this PV_1 is the NPV of a particular realization of future cash flows that is generated in the simulation, and therefore the calculated standard deviation is the outcome of all future uncertainties (Smith 2005). Therefore, the approach overestimates the volatility. Several alternative approaches have been presented to overcome this issue (see e.g. Brandao et al. 2005a, 2005b; Godinho 2006; Haahtela 2010).

There is another very effortless method for finding the volatility. It is based on the assumptions and qualities of the gBm and its lognormal underlying asset value distribution. Very similar thoughts are presented by Smith (2005) suggesting that correct parameterization for the mean value and volatility could be found by changing the volatility until the underlying asset mean and standard deviation match the simulated cash flow and its standard deviation. However, if common gBm assumptions hold, this can actually be solved

¹ For simplicity, we assume that the volatility is changing only according to the level of the underlying asset price, i.e. it is not time varying. However, taking this into account is very simple by combining the results of this paper with the method of Haahtela (2010).

analytically. Given that PV_0 is known, and it is possible to simulate future cash flows, true or risk-neutral distribution of the cash flows in future can be simulated. As well as discounting all the cash flows to the present value, they can also be undiscounted to their future value. Because the present value of cash flows is known (PV_0) as well as their simulated undiscounted future value with standard deviation, it is possible to find the volatility parameter analytically without any unnecessary additional computations and simulations. It is known that in a risk neutral world, asset value increases with time according to Equation (1), and that the standard deviation of the process increases according to Equation (2).

$$S_t = S_0 e^{\mu t} \quad (1)$$

$$std(S_t) = S_0 e^{\mu t} \sqrt{e^{\sigma^2 t} - 1} = S_0 e^{\mu t} \sqrt{e^{\sum \sigma_i^2 t_i} - 1} \quad (2)$$

Therefore, if the standard deviation of the underlying asset process at certain time point is known, it is possible to compute the average volatility for each time period according to Equation (3) as

$$\sigma_i = \sqrt{\frac{\ln \left[\left(\frac{std(S)_i}{S_0 e^{\mu t}} \right)^2 + 1 \right] - \sum_{i=0}^{i-1} (\sigma_i^2 t_i)}{t_i}} \quad (3)$$

The information required is the length of time period t for volatility estimation, value of asset S_0 at the beginning, interest rate r , and standard deviation $std(S)$ of the asset value at the end of volatility estimation period. Interest rate μ does not change the volatility estimation as long as the same interest rate is used both in cash flow simulation and when computing the volatility. Therefore, even if the risky expected return is used in volatility estimation, the option valuation still follows risk-neutral pricing with risk-free interest rate used.

4. Stochastic underlying asset processes

Basic financial options models assume that the dynamic underlying asset value process is geometric, i.e. multiplicative. Therefore, the underlying asset (stock price or project value) distribution is lognormal. The properties of lognormal distribution are that the value cannot fall below zero, but it may increase to infinity without an upper limit. The distribution is positively skewed, having most of the values closer to the lower limit, and the natural

logarithm of the distribution yields to normal distribution. Therefore, Black and Scholes (1973) and many other option valuation methods make the assumption of lognormal underlying asset value with a normally distributed rate or return distribution.

While some assets can randomly fluctuate according to Brownian motion, prices of raw materials and commodities tend to fluctuate randomly around a certain normal value of the asset, that is, the marginal cost of producing the commodity. Such processes are mean reverting. The Ornstein-Uhlenbeck process is the best known of these. A disadvantage of the mean-reverting models is that often they cannot be solved analytically, although numerical methods are sufficiently effective nowadays for solving the values. However, numerical techniques are otherwise tedious to model. Furthermore, parameterizing the models is often hard, even with historical data available. Another shortcoming of some models is the possibility of negative underlying asset values.

Another alternative for estimating price changes over time is to assume that the price changes are additive, meaning that the price between times t and $t + 1$ will either rise or fall by a predefined amount with different risk-neutral probabilities. Additive models are simple to use, but they lack realism if used for the valuation of financial options. Stock price changes are rather geometric than arithmetic. Secondly, stock prices can never be negative, whereas additive models, if not otherwise restricted, make this possible. However, additive models are still useful for localized analyses, over short time periods, and are a good building block for other models.

The earlier-mentioned multiplicative, mean-reverting, and additive models assume a sufficiently continuous fluctuation and a stream of available information. However, in the context of R&D projects, new knowledge may only arrive at discrete points in time, causing sudden up or down jumps; at other times the changes will probably be quite small. Therefore, Lint and Pennings (1999) and Willmer (1995) suggest using the pure jump diffusion model in pricing real options. Schwartz and Moon (2001) developed their model based on Pindyck (1993) for multi-staged investments with properties of both jump process and continuous geometric Brownian motion. The jump process in various real options models is often modeled with Poisson distributions. Although the models are not mathematically impossible to understand, it may sometimes be challenging for practitioners to find or guess the right

parameterization for them. As Paxson (2003) stated, parameterizing many R&D real option models on an empirical basis is often quite difficult.

The observed distributions of estimated and simulated NPV calculations do not often resemble lognormal distribution but are rather something between normal and lognormal. As a result, a better approximation for the underlying value distribution would be something between the normal and lognormal distribution, indicating skewness and kurtosis in the rate of return distribution. This is not very surprising, given the nature of cash flow calculations. This is mostly because cash flow simulation follows the law of large numbers. According to the central limit theorem, the sum of a sufficiently large number of independent random variables of fairly finite mean and variance is approximately normally distributed. If the process is multiplicative, i.e. the random variables are multiplied, the result is lognormally distributed. Typical cash flow calculation has both of these properties, i.e. they are sums of several cash flow variables, with some of these variables having multiplicative correlated properties over the time periods. Therefore, when we sum up a large number of cash flows over multiple simulation runs, the terminal value distribution is often very close to the shape of something between a normal and lognormal distribution.

Some of those who apply the real options approach state it does not matter if the underlying asset distribution or timeliness behavior of the dynamic price process does not match the often assumed multiplicative assumption, because the financial markets are not too strict about that. Presumably, these statements are based on the assumption of the possibility of constructing a dynamic tracking portfolio. With the option value mimicked exactly by the portfolio, the form of uncertainty doesn't affect the option value (Amram and Kulatilaka 1998). However, applying real option valuation without such tracking and relying on standard deviation as a measure of volatility may misguide decision making. Results of Laughton and Jacoby (1993), Boyarchenko and Levendorskii (2000), and Trojanowska and Kort (2005) show that investment decisions with values and investment thresholds may change significantly if Gaussian distribution is replaced with a non-Gaussian distribution having the same variance.

Cox and Ross (1976) suggested the constant elasticity of variance to describe financial assets dynamics for such cases where volatility decreases as the underlying asset value increases. The rationale for the model is that all firms have fixed costs to meet, regardless of the firm's

operating performance, and therefore if operating performance weakens, volatility increases. Market-observed implied volatilities also support this theory and model. Therefore, constant elasticity model price movements may be considered to have qualities of both the additive and multiplicative model.

Another common problematic assumption is that the underlying asset has only positive values, which is always true for stocks. Unfortunately, this doesn't hold for all real investments, because the operating cash flows – also after the initial stage investment outlays – may still be negative. This does not happen very often, but it is still possible, in which case negative underlying values can cause problems in volatility estimation and option valuation. In such cases, the additive process can be used, as well as Rhys and Tippett's (2003) analytical solution for generalized Student's distributions, which also allows negative underlying values.

The models that build on the assumption of geometric Brownian motion (gBm) could give erroneous results if applied to such cases if the actual outcome distribution and the process causing it did not follow gBm properties. Negative underlying asset values, skewness, and kurtosis, for example, could change the valuation significantly. Therefore, the mean and standard deviation of the underlying asset outcome distribution or the mean and volatility of the rate of return distribution are not always sufficient for describing a sufficiently smooth dynamic process without jumps or mean-reverting behavior. Figure 2 illustrates this with different project outcome distributions. In earlier research, Rubinstein (1983) and Brigo et al. (2004) use the displaced model so that the underlying asset (continuous black line) always has only positive values. Camara (2002) considers also the possibility of having negative underlying asset values, but the underlying asset distribution is right-tailed. The approach and stochastic process suitable for real option applications that handles any of the given distribution shapes in Figure 2 is presented in the following section 5.

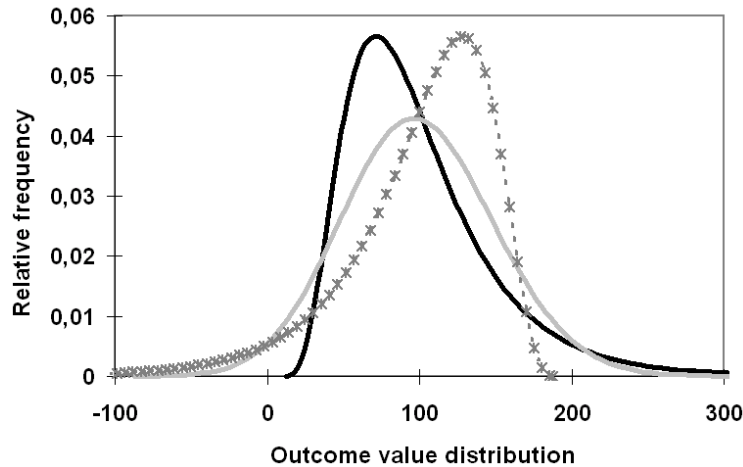


Figure 2: Project outcome distributions with a mean of 100 and standard deviation of 50. Applying methods assuming gBm to other than dynamic process leading into lognormal underlying asset outcome value distribution (the black line in picture) may cause erroneous results.

5. Generalized displaced diffusion process for real options

One type of stochastic process that has not been commonly applied to real options valuation despite its good qualities and popularity in modeling of financial derivatives is the displaced diffusion process of Rubinstein (1983):

$$dS = \mu S dt + \sigma_d (S + \theta) dW \quad (4)$$

The process is extremely easy to solve and numerically fast. The process also allows good approximation for square root process as well as a certain range of constant elasticity of variance processes and extreme value processes (Jäckel 2002). According to the original use of Rubinstein (1983), some of the firm's assets are riskless, whereas others are risk bearing. The displaced diffusion formula and the calculation model consider a firm's assets as a portfolio of risky and riskless assets assuming that the assets of the firm can be decomposed into relatively riskless and risky return portions. The riskless part of the asset value increases each period by risk-free interest rate and the risky part follows a normally distributed random variable process with instantaneous volatility. Therefore, the riskless part is also the lower level for the underlying asset value.

Compared with the original presentation of Rubinstein (1983), we change the process and its idea slightly by taking the shifting parameter θ out of the stochastic part of the process. The shifting parameter does not either have the same meaning as in Rubinstein (1983), and

therefore it has risky expected return μ that is assumed to be non-stochastic. Therefore, stochastic process suggested has the evolution of the underlying asset S out to a time horizon T given by equation (5) as

$$S_T = S_{\theta,0} e^{(\mu - \frac{1}{2}\sigma_d^2)T + \sigma_d \sqrt{T}z} + \theta_0 e^{\mu T} \quad \text{with } z \sim \mathcal{N}(0, 1) \quad (5)$$

where S_{θ} is the displaced mean parameter, σ_d is volatility of the displaced diffusion and θ is shifting parameter. The corresponding underlying asset value distribution is a shifted lognormal distribution. This process is also very similar to the D-binomial process of Camara and Chung (2006) and has been used also in Haahtela (2006). Technically the displaced process presented suggested here consists of two parts: the first part S_{θ} is stochastic and follows ordinary geometric Brownian motion while the other part θ is deterministic. As a result, the expected value and standard deviation of the underlying asset S at horizon T are given as

$$S_t = (S_{\theta,0} + \theta_0) e^{\mu t} \quad (6)$$

$$std(S_t) = |S_{\theta,0}| e^{\mu t} \sqrt{e^{\sigma_d^2 t} - 1} \quad (7)$$

Equations (6) and (7) reveal the similarity with the ordinary geometric Brownian motion. The absolute value of S_{θ} is taken because this parameter may also get negative values². The corresponding underlying asset value distribution for the process (5) is a displaced (shifted) lognormal distribution. We will later use notation $(S_{\theta}; \theta; \sigma_d)$ to describe the process and the resulting distribution following equation (5). Displaced diffusion process with different values of S_{θ} and θ is capable to model stochastic processes that are between multiplicative (lognormal) and additive (arithmetic) processes. It provides also good approximation for certain extreme value processes, and it also allows negative underlying asset values.

The parameterization of the approach based on a cash flow simulation is very simple and a straightforward extension of the generalized risk-neutral volatility estimation. After the cash flow calculation simulation, we fit the present value at time zero. Most simulation software have the possibility to fit the simulated values to the displaced diffusion process. However, because equation (7) assumes the standard deviation of the terminal value distribution in

² Although this is very unlikely to happen with any real-life case.

future, the standard deviation parameter given by the distribution-fitting software is multiplied by e^{rt} . Then, parameters S_0 , θ and standard deviation at maturity can be used to calculate the displaced volatility σ_d according to equation (7).

6. Displaced binomial tree

Lattice valuation models are based on a simple representation of the evolution of the underlying asset value. The two main ideas with lattice approaches are: 1) the modeling of the continuous process with a discrete random walk, and 2) the assumption of risk-neutral pricing (Wilmott et al., 1995, 180-181). In the continuous limit, a lattice with an infinite number of time steps to expiration represents a continuous risk-neutral evolution of the asset value. Lattice models are accurate, robust, and intuitively appealing tools for valuing financial and real options (Hahn 2005, p. 6). Lattices are much more easily explained to, and accepted by, management because the methodology is much easier to understand (Mun 2006). This is valuable, especially with sequential and parallel compound options.

In a lattice method, a tree of possible values of underlying asset prices and their probabilities, given an initial asset price, is built. This tree determines the possible asset prices and the associated probabilities of these asset prices being realized. In other words, a lattice determines the asset's prices and probabilities in a state space during each time period over the security's lifetime and at expiry of the security. The possible values of the security and therefore also the payoff of the option at expiry can then be calculated, and finally, by working back down the tree, the security can be valued (Wilmott et al. 1995, 182). The most simple presentation of a lattice model is a binomial model. A binomial approximation for the geometric Brownian motion process may be developed by assuming that during a short time interval Δt , stock prices jump from an initial value, S , either up to the new value, S_u , or down to the new value, S_d . The transition probability of moving up to S_u is assumed to be p , so the probability of moving down to S_d is $1-p$.

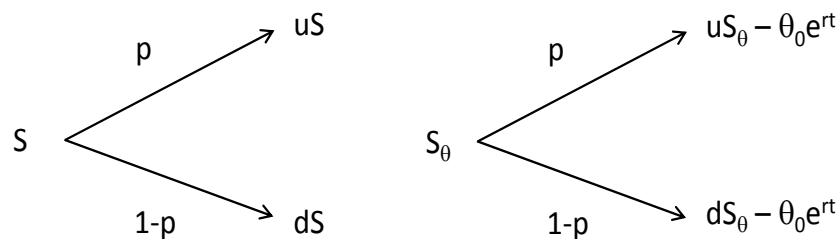


Figure 3: Binomial time step assuming the stochastic process follows geometric Brownian motion (left) and the displaced geometric diffusion process (right).

These parameters are chosen so they give correct values according to the continuous-time process for the mean and variance of the change in the stock price during the time interval Δt . The following conditions are sufficient to ensure this convergence:

- a) Jumps are independent of the stock price level
- b) The mean of the binomial distribution is equal to the mean of the lognormal distribution
- c) The variance of the binomial distribution is equal to the variance of the lognormal distribution
- d) The probabilities p_u and p_d are positive and between 0 and 1 but not equal to either 0 or 1.
- e) The probabilities sum to 1

$$p_u u + p_d d = e^{r\Delta t} \quad (8)$$

$$p_u u^2 + p_d d^2 = e^{(2r+\sigma^2)\Delta t} \quad (9)$$

$$p_u + p_d = 1; \quad 0 < p < 1 \quad (10)$$

The discretized dynamic process must give correct values to [the] mean, increasing by the risk-free interest rate according to the risk-neutral assumption (Equation 8), and variance of the asset dynamics at each time period of length Δt (Equation 9). Equation (10) ensures that the transition probabilities remain between 0 and 1, a necessary condition for the discrete world represented by the tree to preclude arbitrage. Another common restriction is the recombination condition $u \cdot d = d \cdot u$ so that the binomial lattice branches reconnect at each step.

The best known example of the binomial tree parameterization is Cox, Ross and Rubinstein (1979), which is also often used in context of real options. Their suggestion is an exact solution to the mean value but only an approximation of variance. CRR also loses stability if $\Delta t > \sigma^2/r^2$ and, as a result, the other probability becomes larger than one and another smaller than zero. This risk is relevant in the case of real options because the time steps Δt used are often longer than those employed with financial options. This is often stated not to be a problem because the numerical accuracy required is not as significant as it is with financial options. This statement, however, neglects that the lattice construction may not be

valid anymore. Secondly, and even more importantly, the volatility parameter in case of displaced diffusion (displaced volatility) may become very small if the terminal value distribution of the cash flows is very close to normal distribution. Therefore, another alternative is suggested for CRR.

There are several ways to define a binomial tree. Chance (2007) discusses 11 different alternatives according to their properties, theoretical correctness, use of either raw mean and variance or log mean or variance etc. The majority of the anomalies occur only with a combination of very extreme values and a small number of time steps. They can usually be avoided when actually used for pricing options. The problems mostly occur when the models are used for illustrative purposes with a small number of time steps. Two good qualities are that the model recovers σ for finite N and prohibits arbitrage for finite N. Also, robustness against several anomalies (e.g., risk-neutral probability $0 < p < 1$; $u > 1$; $d < 1$; $ud > 0$) with good convergence to the continuous time value is a preferable quality. Based on these qualities, parameterization of Wilmott et al. (1995) is chosen. It uses both raw mean and raw variance for the stochastic process description (instead of their log versions), recovers volatility correctly without arbitrage for a finite number of time steps, and is very robust against anomalies in practice. The risk-neutral probabilities for the up and down movements are $p_u = p_d = 1/2$, and then the corresponding sizes for up (u) and down (d) movements are given as follows:

$$u = e^{r\Delta t} \left(1 + \sqrt{e^{\sigma^2 \Delta t} - 1} \right) \quad (11)$$

$$d = e^{r\Delta t} \left(1 - \sqrt{e^{\sigma^2 \Delta t} - 1} \right) \quad (12)$$

Another advantage is that this binomial tree is also suitable for displacing. The only difference is that instead of using S and σ , the displaced S (S_θ) with corresponding σ_θ are used instead, and the shifting parameter is subtracted from the value.

As an illustration, we choose a straightforward four-year example. The project's underlying asset value S at the beginning is 100. Then, after four years, the company has the option to leave the project for a salvage value of 50. The annual volatility σ in the first case, assuming geometric Brownian motion, is 0.5 and the standard deviation at maturity is 160.1. In case of the displaced diffusion process, the expected value is also 100 and the standard deviation is

160.1, but the stochastic process is defined so that S_0 is 1000, the threshold parameter θ is -900, and the displaced volatility σ_d is 0.0653. As a result, we construct two different binomial trees following processes (100; 0; 0.5) and (1000; -900; 0.0653).

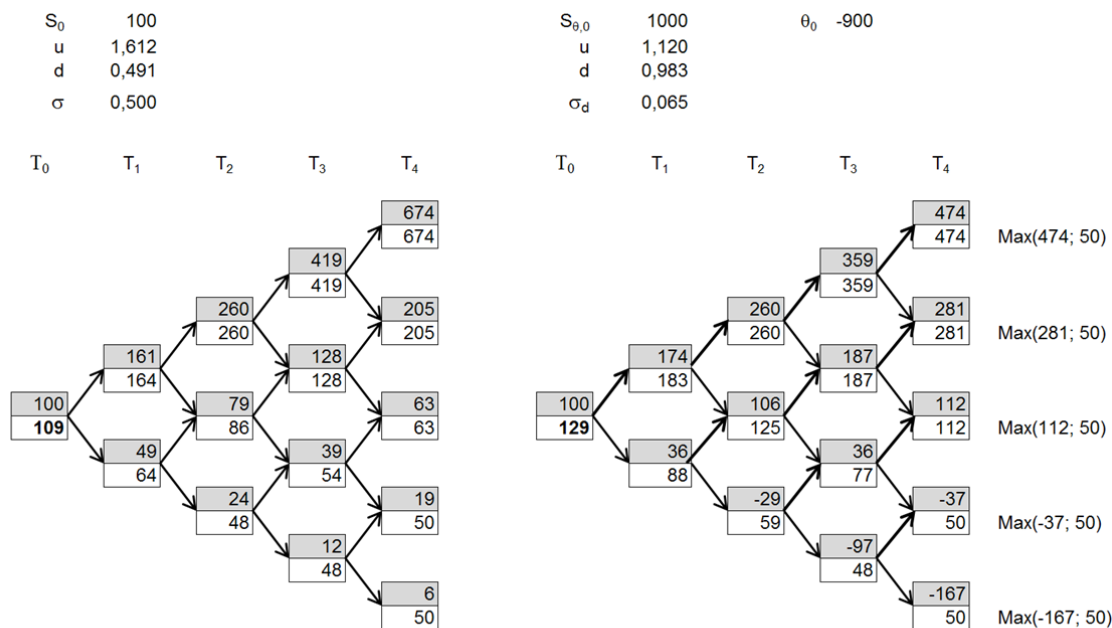
First, both lattices are constructed starting with the S and S_0 . The underlying asset value goes either up or down during each time period. These discretized values of the underlying asset processes are presented with a gray background color in the binomial lattices (see Figure 4). For example, in case of the displaced binomial lattice, the first up state value after the initial node is calculated as $1000 * 1,120 - 900 * \exp(0.05*1) = 1120 - 946 = 174$. This process is described on a gray background in the binomial lattices.

After the underlying asset value paths are constructed, the option payoff functions V are entered into the model. In this case, the option is exercisable only at maturity and the payoff function for each end node is $V = \text{Max}(S; X)$ ³. Then, the option tree is evaluated by starting at the end of the tree and working backwards with dynamic programming according to the risk-neutral valuation. Because of the risk-neutrality assumption, the value at each inner node can be calculated as the expected value at time t discounted at rate r for a time period V_{t-1} according to:

$$V_{t-1} = \frac{p_u V_{t,u} + p_d V_{t,d}}{e^{r\Delta t}} \quad (13)$$

The option value at each state is given in the picture on the white background. For example, the value of 77 (in Figure 4, on the right displaced binomial tree) is calculated as $(0.5*112 + 0.5*50)/(\exp(1*0,05)) \sim 77$. Finally, working back through all the nodes provides the value of the option at time zero. The value of the project with the option to abandon for 50 is therefore worth 109, i.e. the option increases the project value by 9. However, if we assume the same underlying asset value and even the same standard deviation at maturity, but have a slightly different stochastic process and shape of the terminal value distribution, the project value with the option is 129, so the option value is 29. This difference of 29 vs. 9 can be considered significant.

³ This is different from the most common option payoff functions, which are for the call option $\text{MAX}(S-X; 0)$ and for the put option $\text{MAX}(X-S; 0)$. For simplicity, we also assume that there are no acquiring costs in this example.



In both cases, $T = 4$, $n = 4$, $\Delta t = 1$, $r_f = 5,0\%$, $p = (1 - p) = 0.5$, $\text{Std}(S_\theta) = 160.1$, $X_a = 50$

Figure 4: Comparison of the ordinary gBm assuming lattice (left) and the displaced diffusion binomial lattice (right). Lattices use the parameterization of Wilmott et al. (1995) for up and down movements.

This example illustrates three strengths of the displaced diffusion process. Firstly, the proportional level of uncertainty changes with the level of the underlying asset. In the event of going up twice on both trees during time period 2, the underlying asset value is 260. During the following time step, the left binomial tree assuming common geometric Brownian motion, either increases to 419 or decreases to 128, while the displaced diffusion tree only changes to either 359 or 187. Because both trees have the same risk-neutral probabilities of 0.5 for both up and down movements⁴, we can see that the displaced diffusion tree with these parameters has a smaller proportional deviation for higher values. On the other hand, if we take two steps down in the beginning, the lognormal tree has a difference of only 27 ($39 - 12$) between the up and down movements, while the displaced binomial tree has a difference of 133 ($36 - (-97)$).

The second important property is that the underlying asset may also have negative values, but this does not cause difficulties to the valuation approach. Many approaches, relying on taking the natural logarithm during volatility estimation, σ , often fail in case of negative numbers.

⁴ Even if these are risk-neutral probabilities and not real probabilities, these probabilities are related (see e.g. Amram and Kulatilaka 1998, p. 118, or Hull 1997)

The third important feature is that the case also illustrates how to investigate what is the maximum loss without the option to abandon. According to the lognormal process assumption, the project without options has a value of 6 to the company in the worst case. However, if we assume the displaced diffusion process, this worst case value for the project is -167. This difference may be very significant to the decision maker. Therefore, assuming the lognormal process may also hide the true downside risk of the project without options while the displaced diffusion process does not do that.

7. The case example

After the illustration of the displaced diffusion binomial tree construction and its properties we investigate what the impact of the displaced diffusion is instead of using the common lognormal process. Therefore, we chose a case study, and set its expected value and standard deviation as constant, but the stochastic process form and the distribution shape was altered. Also, standard deviation (volatility) was changed to compare how much the changes in the stochastic process form affect the option values in comparison with the change in volatility.

The case study chosen was originally presented in Trigeorgis (1991). Later, the same case was used in Trigeorgis (1996), in a collection of articles on real options by Schwartz and Trigeorgis (1998) and also by Benaroch (2001). Therefore, researchers in the field are most likely familiar with this specific case. Secondly, objectivity is better maintained as the researcher does not have the opportunity to manipulate the case example according to his own needs⁵.

⁵ And there is no need to reveal or disguise real company data that was originally used for investigating the topic.

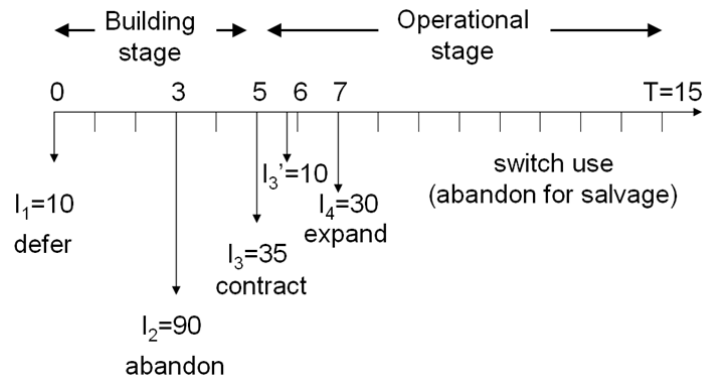


Figure 5: Cash flow structure of the case study for comparing displaced diffusion binomial lattice properties with ordinary gBm assuming lattice.

The case example is illustrated in Figure 5. The project consists of two stages, the building stage lasting five years, and the operational stage, which lasts ten years. Cash inflows are assumed to occur only after the building stage is finished. The initial payment for the investment is I_2 . However, the company may also wait up to two years until starting the construction. Three years after the construction, the company has to invest in I_2 to continue the project. The company may also abandon the project at this stage, losing only the initial investment of I_1 . If the company continues construction, it will most likely invest in I_3 in year five. However, if the future outlook isn't that bright, the company may contract this investment by investing in I_3^* , but then it will lose 25% in forthcoming cash inflows. During the operating stage, the company has two alternatives besides doing business as usual. Firstly, the project can be expanded with an investment outlay of I_4 . This is assumed to increase the project cash inflows by 35%. However, the company may also abandon the project for its salvage value at any time during the operating stage. However, the salvage value erodes continuously as time passes.

The following bulletin point list summarizes the information needed for the valuation of the project and its options.

- The initial present value, also called the gross present value, V , is 100 in the beginning
- The annual risk-free rate, r , is 5 %
- The annual volatility of the project's value, σ , is 0.5.
- The opportunity to invest is deferrable by two years until $T_1 = 2$

- The initial investment outlay I_1 is 10
- The second investment $I_2 = 90$, happens at T_2 , three years after the initial investment. However, the construction can be abandoned without recovery by forgoing the second-stage investment.
- The third investment $I_3 = 35$ is done five years after the initial investment in time T_3 , given that the project was not abandoned in the earlier stage. The normal investment can also be contracted by 25% by reducing the third investment I_3 to $I_3^* = 10$.
- The scale of the project can be expanded by 50% seven years after the initial investment at time $T_4 = 7$ by making an additional investment $I_4 = 30$
- The project can be given up at any time during the operational stage (after T_3). The project's salvage value is the cumulative value of investments declining exponentially at an annual rate of 10%.

The construction of the lattices was made in Excel, with monthly time steps. The results of the study are presented in the following.

8. Case results

Table 1 presents the extended NPV, the total value of the project, with all 21 possible combinations of the five options, with the standard deviation equaling the volatility of 0.5. The letters at the left describe the combination of options with the meanings (D)efer, (A)bandon, (C)ontract, (E)xpand, and S(witch). All 12 parameterizations are such that the underlying asset distributions have the same expected value and standard deviation (1364). The models differ from each other only according to the stochastic process leading to the terminal value distributions. Figure 6 illustrates the differences between distribution shapes with different skewness and kurtosis for all cases where S_0 is positive.

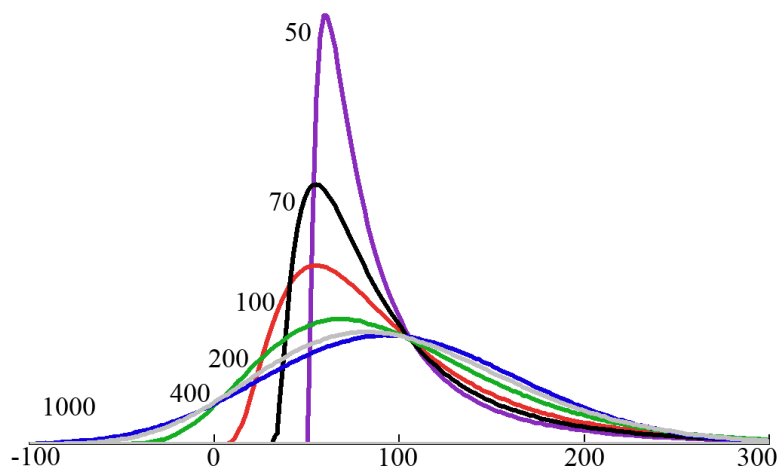


Figure 6: Illustration of different distribution shapes with the same mean and standard deviation. Cases with negative S_0 are horizontal mirror images of these distributions.

The smaller the S_0 , the more the underlying asset value distribution and, therefore, also the rate of return distribution has peakness. Cases with S_0 of 50 and 70 have more peakness than the usually assumed case of lognormal distribution. The higher the S_0 , the more the underlying asset distribution resembles normal distribution. The cases with negative S_0 are horizontal mirror images of these distributions.

Table 1: Comparison of the project values based on the 12 different displaced diffusion binomial lattice results, with all having the same standard deviation and mean but different stochastic processes.

CASE	A	B	C	D	E	F	G	H	I	J	K	L
S_0	50	-50	70	-70	100	-100	200	-200	400	-400	1000	-1000
θ	50	150	30	170	0	200	-100	300	-300	500	-900	1100
σ_0	0,584	0,584	0,545	0,545	0,500	0,500	0,403	0,403	0,292	0,292	0,152	0,152
Std(S)	1364	1364	1364	1364	1364	1364	1364	1364	1364	1364	1364	1364
D = defer	15,0	13,8	19,9	18,9	26,3	25,6	43,8	43,2	64,7	64,1	84,6	83,9
A = abandon	8,2	6,4	14,4	12,6	22,5	20,7	43,2	41,7	68,3	67,7	93,4	92,8
C = contract	-12,3	-9,7	-10,4	-7,9	-7,7	-5,5	-0,8	1,1	7,5	9,1	16,1	16,7
E = expand	14,1	24,0	15,2	27,9	20,2	33,2	36,1	47,7	56,8	65,5	78,1	82,8
S = switch	2,9	16,6	11,7	27,8	25,0	43,2	65,6	85,5	120,8	138,6	180,9	192,0
DA	19,5	19,7	26,8	27,4	36,7	37,7	63,7	64,5	96,2	97,3	129,1	129,4
DC	15,3	14,7	20,5	20,2	28,0	27,5	47,5	46,5	70,3	70,0	92,3	93,3
DE	36,2	48,5	43,1	56,6	54,6	67,5	86,7	98,1	126,3	134,5	166,0	168,9
DS	18,0	29,4	26,2	41,3	39,1	57,7	81,0	102,8	139,7	159,9	205,3	217,2
AC	8,4	6,8	14,7	13,1	23,0	21,6	44,2	43,4	70,9	69,6	96,3	96,2
AE	28,4	39,3	37,6	48,7	50,7	61,2	85,9	94,6	128,1	135,5	171,1	173,7
AS	9,9	16,7	18,0	27,9	30,0	43,3	68,2	85,5	121,9	138,6	181,2	192,0
CE	15,5	28,6	18,9	34,1	26,5	41,9	49,2	62,6	78,0	87,8	107,1	112,7
CS	2,9	16,6	11,7	27,8	25,0	43,2	65,6	85,5	120,8	138,6	180,9	192,0
ES	23,1	44,2	35,6	56,6	53,5	73,5	101,6	118,8	161,0	174,3	222,1	229,1
DAC	19,5	19,9	26,9	27,8	37,2	38,3	64,7	65,6	97,7	99,4	131,7	131,6
DAE	41,1	58,0	52,2	69,6	68,4	85,1	112,0	126,5	166,4	176,2	219,9	226,4
DAS	20,3	29,5	28,7	41,4	41,9	57,8	83,3	102,8	141,1	159,9	205,8	217,2
DCE	36,4	50,6	44,5	59,7	57,1	72,0	92,3	104,0	135,1	144,0	177,2	183,4
DCS	18,2	29,5	26,4	41,3	39,3	57,8	81,3	102,8	139,9	159,9	205,5	217,2
DES	35,0	68,1	48,9	83,4	69,4	104,4	128,3	161,9	205,8	233,8	289,8	305,5
ACE	28,5	39,8	38,0	49,6	51,5	62,7	87,5	96,6	131,2	138,8	176,1	179,7
ACS	9,9	16,7	18,0	27,9	30,0	43,3	68,2	85,5	121,9	138,6	181,2	192,0
AES	28,6	44,4	40,7	56,8	57,7	73,7	104,4	119,0	162,7	174,6	223,0	229,4
CES	25,6	47,4	38,1	59,7	56,0	76,6	103,9	121,7	163,7	176,7	223,7	230,8
DACE	41,1	58,7	52,4	70,2	68,8	86,2	113,7	128,0	168,7	179,7	224,3	230,2
DACS	20,4	29,5	28,9	41,4	42,2	57,8	83,7	102,9	141,5	160,0	206,2	217,3
DAES	41,8	76,1	55,3	93,7	75,7	118,1	141,4	185,0	231,8	269,6	332,0	354,2
DCES	39,2	76,1	52,6	93,7	72,7	118,1	139,2	185,0	230,6	269,6	331,6	354,1
ACES	30,4	47,6	42,5	59,9	59,6	76,8	106,2	121,9	165,1	176,9	224,4	231,0
DACES	41,8	76,1	55,3	93,7	75,7	118,1	141,4	185,0	231,8	269,6	332,0	354,2

The results show that the values change significantly when the terminal value distribution is not assumed to be lognormal. The base case values (100; 0; 0.5) are presented with a dark gray background in Table 1. With all options (DACES), the base value of 75.7 changes between the minimum of 41.8 and the maximum of 354.2 according to the shape of the distribution and the stochastic process. The closer the underlying asset's stochastic process is

to the normal process, the higher the option values. Cases (1000; -900; 0.152) and (-1000; 1100; 0.152) are very close to the normal distribution⁶, and therefore their results are quite close to each other.

The results in Table 1 do not mean that the ordinary lognormal assumption would usually undervalue the project. In the Trigeorgis (1993) case setup, with the parameterizations chosen here, the value of the project clearly increases according to two different underlying asset distribution shape-related dimensions: firstly, the closer the distribution shape is to the normal distribution, the higher the combined option value; and secondly, when the distribution is skewed a little to the right. This example, however, may not be the best alternative. The project is rather long, and the annual volatility of 0.5 is also very large. Had we used smaller volatility 0.3, the base case value of all options in the case of (100; 0; 0.3) would be 50.6, and the maximum value for all the projects with all the options (DACES) is achieved when (-1000; 1100; 0.043) and the corresponding project value is 93.0 (see Table 2).

As is commonly known, the higher the volatility and option time to maturity, the higher the options prices. In this particular case, both of these base-case parameter values are very high, with a volatility of 0.5 and time to maturity of 15 to 17 years. Therefore, the results for a volatility of 0.3 should be considered as realistic rather than the assumption of 0.5 suggested in the original case. The changes in case results with more realistic standard deviations of 0.3 or 0.4 are modest in comparison with the results calculated with higher volatilities⁷. However, this does not mean that the shape effect would be of no interest and would not affect the option valuation. Changes in value from 63.2 to 179.8 or from 50.6 to 93.0 are significant indeed. Also, comparing the original lognormally distributed basic case (100; 0; 0.5) to the case being closer to normal distribution (but still being between normal and a lognormal distribution) and having a much smaller standard deviation of $\sigma = 0.043$, the value of the project in the case of (1000; -900; 0.043) is still higher (88.4 vs. 75.7). Therefore, we can conclude that the stochastic process also has so significant impact on the project value, and therefore we definitely should not settle for the common assumption of gBm when valuing projects.

⁶ Alternative K has a little skewness to the left in comparison with a normal distribution while the alternative L has skewness to the right.

⁷ See the Appendices for the cases where $\sigma = 0.3$ and $\sigma = 0.4$.

Table 2: Comparison of the case values with different stochastic processes (gBm) and displaced diffusion processes with three different volatilities. The results clearly indicate that the form of the stochastic process also has significant impact on the option values.

CASE	volatility 0.5				volatility 0.4				volatility 0.3			
S_0	100	-100	1000	-1000	100	-100	1000	-1000	100	-100	1000	-1000
θ	0	200	-900	1100	0	200	-900	1100	0	200	-900	1100
σ_0	0,50	0,500	0,152	0,152	0,40	0,400	0,080	0,080	0,30	0,300	0,043	0,043
Std(S)	1364	1364	1364	1364	670	670	670	670	358	358	358	358
D = defer	26,3	25,6	84,6	83,9	21,0	20,3	43,5	42,9	15,4	14,8	22,5	22,3
A = abandon	22,5	20,7	93,4	92,8	15,7	14,3	43,3	42,7	8,7	7,9	17,6	17,3
C = contract	-7,7	-5,5	16,1	16,7	-9,5	-7,6	0,3	0,7	-11,2	-9,8	-7,7	-7,6
E = expand	20,2	33,2	78,1	82,8	17,9	28,0	42,3	44,7	15,9	22,6	24,7	25,9
S = switch	25,0	43,2	180,9	192,0	16,1	30,4	78,9	83,6	6,9	16,5	28,0	30,0
DA	36,7	37,7	129,1	129,4	29,1	29,4	65,2	65,6	20,8	21,2	32,5	32,8
DC	28,0	27,5	92,3	93,3	22,1	21,8	47,2	47,6	16,1	15,9	23,8	24,1
CS	25,0	43,2	180,9	192,0	16,1	30,4	78,9	83,6	6,9	16,5	28,0	30,0
ES	53,5	73,5	222,1	229,1	42,5	58,4	112,1	115,6	31,2	43,0	56,3	58,0
DAC	37,2	38,3	131,7	131,6	29,3	29,9	66,1	66,8	21,0	21,4	33,0	33,2
DAE	68,4	85,1	219,9	226,4	58,1	71,1	120,7	123,4	47,4	57,0	69,4	71,3
AES	57,7	73,7	223,0	229,4	46,3	58,8	113,3	116,3	34,6	43,6	57,9	59,3
CES	56,0	76,6	223,7	230,8	45,1	61,7	114,4	118,1	33,9	46,1	59,0	61,0
DACE	68,8	86,2	224,3	230,2	58,4	72,1	122,3	125,5	47,6	57,4	70,4	72,0
DACS	42,2	57,8	206,2	217,3	32,7	44,8	97,6	102,3	23,0	30,4	43,7	45,5
DAES	75,7	118,1	332,0	354,2	63,2	97,7	169,6	179,8	50,6	74,8	88,4	93,0
DCES	72,7	118,1	331,6	354,1	60,8	97,6	168,9	179,5	48,6	74,6	87,4	92,3
ACES	59,6	76,8	224,4	231,0	48,3	61,9	115,4	118,7	36,8	46,6	60,3	62,0
DACES	75,7	118,1	332,0	354,2	63,2	97,7	169,6	179,8	50,6	74,8	88,4	93,0

Other findings in this case setup are very similar to the original findings of Trigeorgis (1993). The option values are non-additive, meaning that a combined value of several options is not the sum of their separate values, but rather smaller than that. This is especially true for options of a similar type, i.e. abandon, switching, and contracting, which are all used all in case of bad project outcomes and which are significantly overlapping. Therefore, the transition from gBm to the displaced diffusion process does not change or extend previous knowledge about the options' interactions.

9. Conclusions

This paper presented a displaced diffusion binomial tree for real option valuation. The method is based on using a displaced stochastic process that 1) has an underlying asset level of changing uncertainty, and 2) allows negative underlying asset values. Also, parameterization for the tree was presented based on a simulated cash flow calculation. The

method is very easy to understand and even more straightforward to use than the original approach of Copeland and Antikarov (2001). Another advantage of the approach is that it reveals the downside risk of cash flows while the commonly used geometric Brownian motion does not allow negative values. For example, the value of the option to abandon the project is much higher if it prevents such losses.

The results of the case with multiple interacting real options show how significantly the value of the project may change as a result of the change in stochastic process form. Especially with higher annual volatilities of 0.5 and 0.4 and with their corresponding displaced volatilities, the shape of the process changed option values strongly, while in the case of a smaller volatility of 0.3, the changes were of smaller magnitude. However, even then they were still significant.

The analysis is based on a single case study, and therefore it is not possible to make strong arguments about how common such significant value changes our results indicate are due to the change in the underlying stochastic process. However, we can conclude that the form of the stochastic process may have such a strong impact on the project value that we definitely should not settle for the common assumption of geometric Brownian motion when valuing projects with real options.

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Appendix I: Option values with annual volatility of 0.3.

CASE	A	B	C	D	E	F	G	H	I	J	K	L
S_0	50	-50	70	-70	100	-100	200	-200	400	-400	1000	-1000
θ	50	150	30	170	0	200	-100	300	-300	500	-900	1100
σ_0	0,410	0,410	0,358	0,358	0,300	0,300	0,190	0,190	0,105	0,105	0,043	0,043
Std(S)	358	358	358	358	358	358	358	358	358	358	358	358
D = defer	10,0	9,3	12,6	12,0	15,4	14,8	20,0	19,4	22,1	21,5	22,5	22,3
A = abandon	2,5	1,1	5,5	4,4	8,7	7,9	14,3	13,7	17,1	16,4	17,6	17,3
C = contract	-13,4	-11,7	-12,4	-10,8	-11,2	-9,8	-9,2	-8,3	-8,1	-7,6	-7,7	-7,6
E = expand	14,1	19,4	14,3	20,9	15,9	22,6	20,3	25,2	23,3	26,1	24,7	25,9
S = switch	-3,4	5,4	1,3	10,7	6,9	16,5	18,2	25,8	25,3	29,9	28,0	30,0
DA	13,0	12,9	16,7	16,8	20,8	21,2	28,1	28,1	31,7	31,9	32,5	32,8
DC	10,1	9,9	13,0	12,7	16,1	15,9	21,0	21,0	23,3	23,5	23,8	24,1
DE	31,4	39,3	34,9	42,9	39,7	46,9	48,6	53,8	53,5	56,8	55,4	56,8
DS	11,7	18,0	15,6	23,6	20,9	29,9	31,9	39,8	39,2	44,2	42,1	44,3
AC	2,5	1,2	5,6	4,6	8,9	8,2	14,7	14,2	17,5	17,1	18,0	17,8
AE	22,1	29,3	26,8	33,5	32,3	38,3	42,2	46,4	47,6	49,8	49,3	50,0
AS	3,2	6,3	7,2	11,5	11,8	17,3	21,4	26,7	27,6	30,9	29,9	31,3
CE	14,6	22,0	15,9	24,3	18,7	26,9	25,1	31,0	29,2	32,5	31,0	32,4
CS	-3,4	5,4	1,3	10,7	6,9	16,5	18,2	25,8	25,3	29,9	28,0	30,0
ES	17,0	30,8	23,4	36,6	31,2	43,0	45,2	53,1	53,2	57,7	56,3	58,0
DAC	13,0	13,1	16,7	16,9	21,0	21,4	28,3	28,6	32,0	32,2	33,0	33,2
DAE	34,3	45,8	40,4	51,2	47,4	57,0	60,0	66,4	67,0	70,8	69,4	71,3
DAS	13,3	18,4	17,5	24,0	22,8	30,4	33,8	40,5	40,8	45,0	43,5	45,3
DCE	31,5	40,7	35,4	44,8	40,6	49,2	50,3	56,3	56,3	59,7	58,6	60,0
DCS	11,8	18,0	15,8	23,7	21,1	29,9	32,1	39,9	39,4	44,3	42,3	44,4
DES	30,4	51,7	37,4	58,5	46,5	66,1	63,9	77,8	74,6	82,8	79,1	82,5
ACE	22,2	29,7	27,0	34,2	32,7	39,0	42,8	47,2	48,3	50,9	50,0	51,3
ACS	3,2	6,3	7,2	11,5	11,8	17,3	21,4	26,7	27,6	30,9	29,9	31,3
AES	21,1	31,3	27,2	37,2	34,6	43,6	47,7	54,0	55,2	58,8	57,9	59,3
CES	19,7	34,1	26,1	39,8	33,9	46,1	48,1	56,2	56,2	60,7	59,0	61,0
DACE	34,3	46,1	40,4	51,7	47,6	57,4	60,5	67,2	67,5	71,4	70,4	72,0
DACS	13,3	18,4	17,6	24,1	23,0	30,4	34,0	40,6	41,0	45,2	43,7	45,5
DAES	34,7	58,0	41,8	66,0	50,6	74,8	69,8	88,5	82,7	93,8	88,4	93,0
DCES	33,0	57,9	39,7	65,9	48,6	74,6	68,2	88,2	81,5	93,4	87,4	92,3
ACES	22,9	34,5	29,2	40,2	36,8	46,6	50,2	56,8	57,8	61,5	60,3	62,0
DACES	34,7	58,0	41,8	66,0	50,6	74,8	69,8	88,5	82,7	93,8	88,4	93,0

Appendix II: Option values with annual volatility of 0.4.

CASE	A	B	C	D	E	F	G	H	I	J	K	L
S_0	50	-50	70	-70	100	-100	200	-200	400	-400	1000	-1000
θ	50	150	30	170	0	200	-100	300	-300	500	-900	1100
σ_0	0,498	0,498	0,452	0,452	0,400	0,400	0,289	0,289	0,180	0,180	0,080	0,080
Std(S)	670	670	670	670	670	670	670	670	670	670	670	670
D = defer	12,6	11,7	16,1	15,5	21,0	20,3	31,2	30,6	39,3	38,7	43,5	42,9
A = abandon	5,4	3,7	10,0	8,4	15,7	14,3	27,7	27,1	37,9	37,3	43,3	42,7
C = contract	-12,9	-10,7	-11,4	-9,4	-9,5	-7,6	-5,2	-3,9	-1,8	-0,9	0,3	0,7
E = expand	14,1	21,7	14,7	24,4	17,9	28,0	27,6	35,7	36,6	41,8	42,3	44,7
S = switch	-0,1	11,3	6,6	19,6	16,1	30,4	40,7	54,3	63,7	73,6	78,9	83,6
DA	16,3	16,4	21,9	22,2	29,1	29,4	44,6	45,5	57,9	58,0	65,2	65,6
DC	12,7	12,3	16,8	16,5	22,1	21,8	33,4	33,2	42,0	42,5	47,2	47,6
DE	33,8	44,0	39,2	49,8	47,3	57,3	66,6	74,5	83,0	87,8	92,5	94,3
DS	14,9	24,1	21,0	33,0	30,1	44,5	55,2	70,2	79,9	91,1	96,4	101,7
AC	5,5	4,0	10,2	8,7	15,9	14,7	28,7	27,9	39,1	38,6	44,4	44,5
AE	25,1	34,4	32,1	41,2	41,4	49,7	62,0	69,1	80,0	84,1	90,7	91,8
AS	6,7	11,7	12,7	19,9	21,0	30,6	43,5	54,5	65,2	73,8	79,7	84,0
CE	15,0	25,3	17,3	29,3	22,5	34,4	36,3	45,7	48,6	54,6	56,3	59,1
CS	-0,1	11,3	6,6	19,6	16,1	30,4	40,7	54,3	63,7	73,6	78,9	83,6
ES	20,1	37,6	29,7	46,9	42,5	58,4	71,9	84,1	96,4	104,2	112,1	115,6
DAC	16,3	16,6	22,0	22,6	29,3	29,9	45,2	46,3	58,9	59,1	66,1	66,8
DAE	37,6	52,1	46,4	60,6	58,1	71,1	84,5	94,8	106,8	113,5	120,7	123,4
DAS	16,9	24,3	23,2	33,2	32,5	44,7	57,3	70,5	81,3	91,4	97,3	102,1
DCE	33,9	45,8	40,1	52,4	49,0	60,7	70,0	78,6	87,4	93,7	98,4	101,5
DCS	15,0	24,1	21,2	33,0	30,3	44,6	55,4	70,3	80,0	91,2	96,5	101,8
DES	32,7	60,3	43,2	71,4	58,1	85,9	94,2	117,8	127,4	143,6	149,0	156,4
ACE	25,2	34,9	32,3	42,0	41,8	50,9	63,5	70,4	82,0	86,3	92,7	94,4
ACS	6,7	11,7	12,7	19,9	21,0	30,6	43,5	54,5	65,2	73,8	79,7	84,0
AES	25,0	38,0	34,1	47,3	46,3	58,8	74,4	84,5	98,2	104,8	113,3	116,3
CES	22,6	41,0	32,1	49,9	45,1	61,7	74,3	87,1	99,5	107,6	114,4	118,1
DACE	37,6	52,5	46,6	61,2	58,4	72,1	85,2	96,1	108,4	114,8	122,3	125,5
DACS	16,9	24,3	23,4	33,2	32,7	44,8	57,6	70,5	81,6	91,5	97,6	102,3
DAES	38,3	67,7	48,7	80,8	63,2	97,7	103,9	135,2	143,1	165,3	169,6	179,8
DCES	36,1	67,6	46,3	80,7	60,8	97,6	102,0	135,1	142,0	165,2	168,9	179,5
ACES	26,7	41,2	35,9	50,2	48,3	61,9	76,5	87,4	100,9	108,1	115,4	118,7
DACES	38,3	67,7	48,7	80,8	63,2	97,7	103,9	135,2	143,1	165,3	169,6	179,8